

AeroNetB ASCM Implementation Logbook

5CM506 Data Driven Systems - D-II

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System	AeroNetB Aerospace Supply Chain Management Dashboard
Deployment Mode	Cloud deployment through Render Blueprint, with local fallback for demonstration and testing
Submission Date	May 2026

1. Implementation Details

Layer	Technology	Purpose
Frontend	HTML, CSS, JavaScript	Role-specific web dashboard, charts, search, forms, and CSV export controls.
Backend	Node.js HTTP API	Authentication, RBAC, validation, cross-database joins, audit logging, and static frontend hosting.
Relational DB	PostgreSQL on Render	Structured supplier, part, order, shipment, employee, role, equipment, and audit data.
Document DB	MongoDB Atlas	QC reports, certifications, IoT sensor logs, and compliance flags.
Deployment	Render Blueprint	Creates the web service and PostgreSQL database from <code>render.yaml</code> .

1.1 Access Instructions

Item	Value
GitHub repository	https://github.com/Spyko2001/aeronetb-ascm-dashboard
Render dashboard URL	https://aeronetb-ascm-dashboard.onrender.com

Health endpoint	https://aeronetb-ascm-dashboard.onrender.com/api/health
PostgreSQL connection	Managed by Render through DATABASE_URL; private value is not included in the submission.
MongoDB connection	Managed by Render through MONGODB_URI; private value is not included in the submission.

1.2 Demo Accounts

Role	Auth ID	Password
Procurement Officer	priya.procurement	demo123
Quality Inspector	lena.inspector	demo123
Supply Chain Manager	marcus.manager	demo123
Equipment Engineer	irene.engineer	demo123
Auditor / Regulator	omar.auditor	demo123

2. Refined Logical Models

The first-semester design was continued and refined into a deployable hybrid model. PostgreSQL now stores the structured domains and MongoDB stores flexible document domains. The backend is responsible for validating cross-database references and combining data for the dashboard.

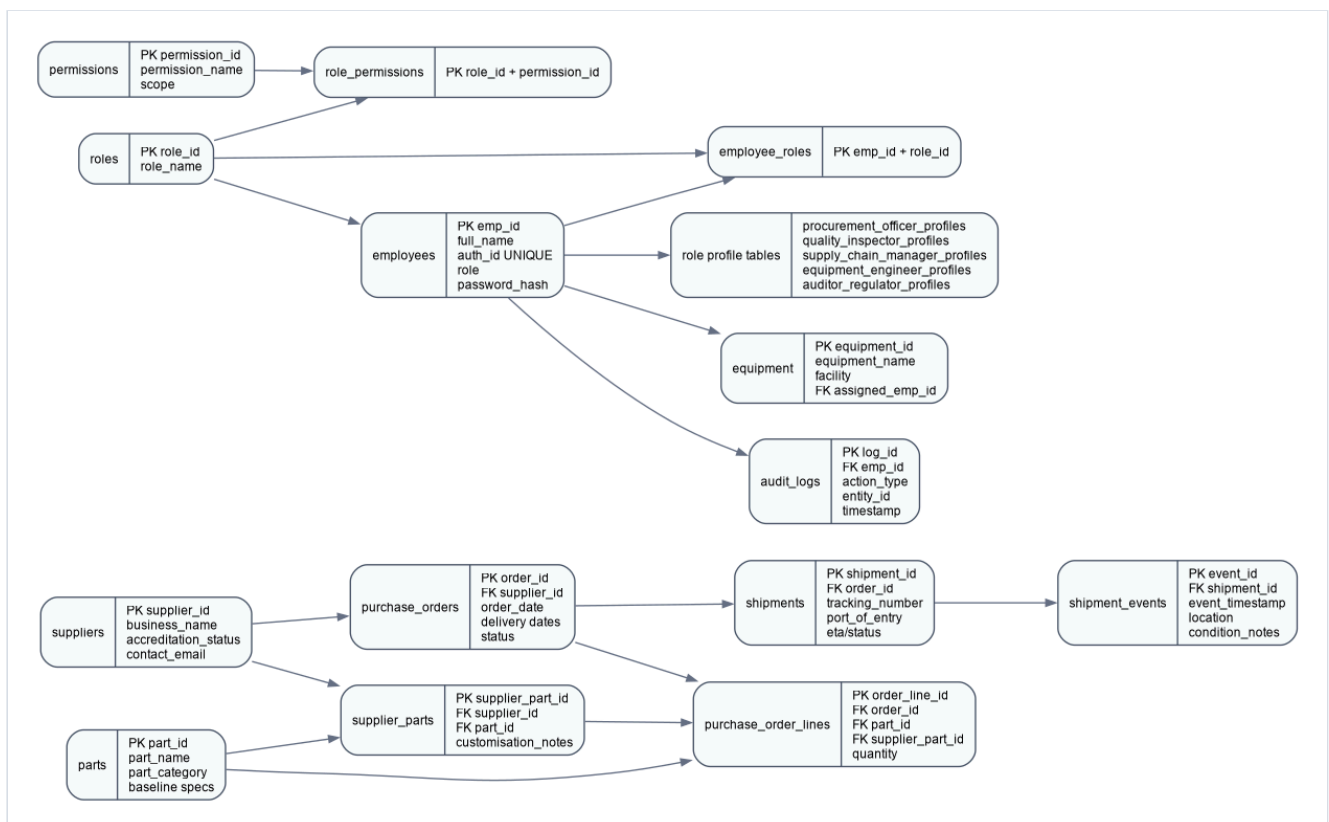


Figure 1 - Refined PostgreSQL relational schema

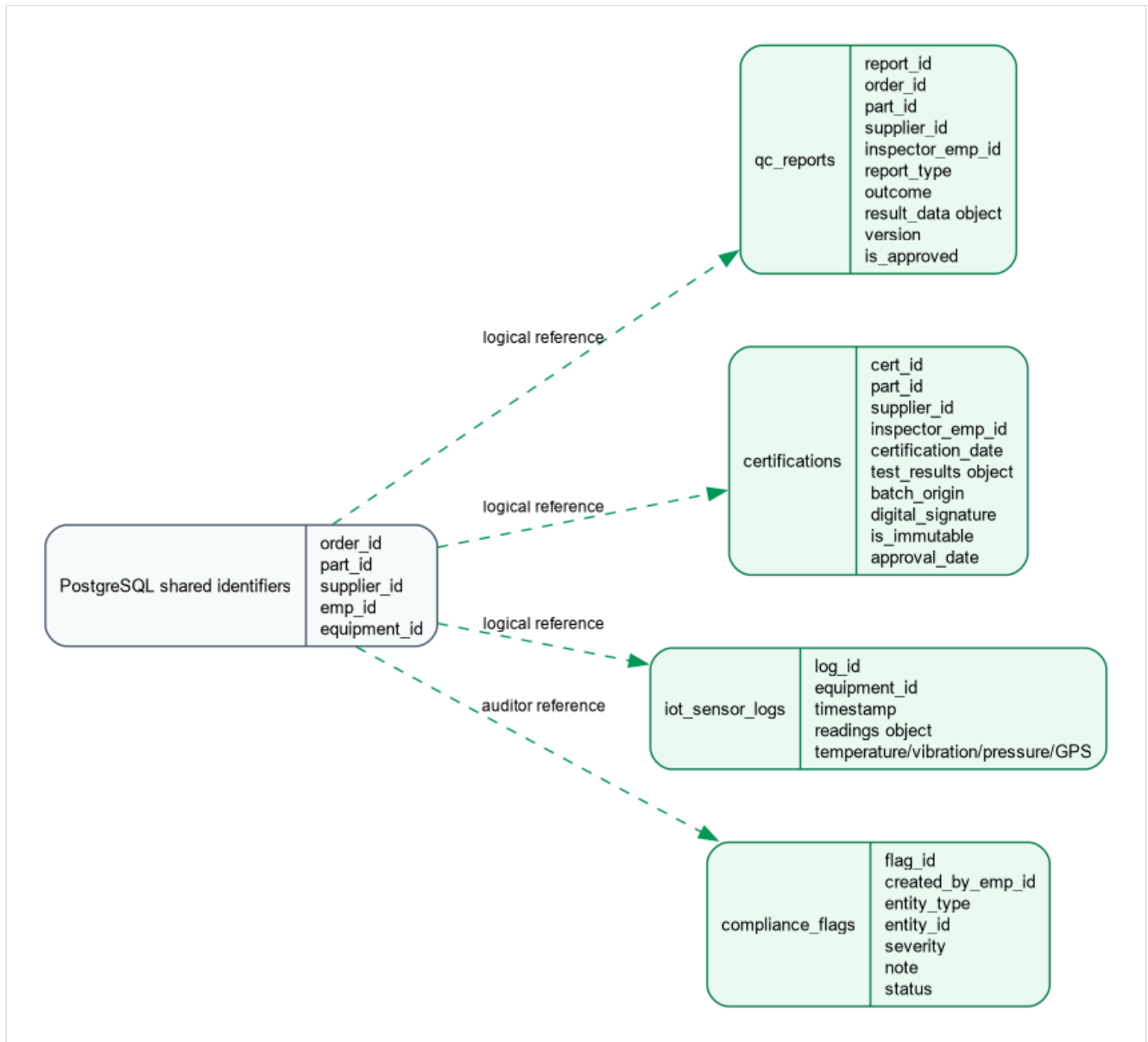


Figure 2 - Refined MongoDB document model

2.1 Refinements From Task 1

Refinement	Implementation
Explicit RBAC model	Added relational roles, permissions, role_permissions, and employee_roles tables, with API checks for every protected endpoint.
Versioned QC reports	New QC submissions create a new MongoDB document version for the same order, part, supplier, and inspection type.
Immutable certifications	The certification update route rejects changes after approval by checking is_immutable.
Audit coverage	Login, view, create, update, approve, export, and flag actions create audit entries with emp_id.
Deployment readiness	Added Render Blueprint, environment variables, automatic schema creation, automatic seeding, and a local memory fallback.

3. Database Implementation

Evidence File	Description
sql_scripts/ddl.sql	PostgreSQL table definitions, keys, constraints, and indexes.
sql_scripts/dml.sql	Dummy relational data for suppliers, parts, orders, shipments, employees, roles, profiles, equipment, and audit logs.
mongo_scripts/collections.js	MongoDB collection creation, JSON Schema validators, and indexes.
backend/src/seed-data.js	Runtime seed records used by the Node application for PostgreSQL and MongoDB bootstrap.

4. API Design and Development

Method	Endpoint URI	Description	Data Source	Role Access
POST	/api/auth/login	Authenticates a seeded user and returns a signed token.	PostgreSQL employees	Public with valid credentials
GET	/api/dashboard	Returns KPIs, alerts, supplier performance, QC summary, shipments, and IoT status.	PostgreSQL + MongoDB	All roles
GET / POST	/api/suppliers	Lists suppliers or creates a supplier.	PostgreSQL suppliers	Read: all relevant roles; write: procurement
GET / POST / PATCH	/api/orders	Lists, creates, or updates purchase orders.	PostgreSQL orders and order lines	Read: procurement/manager/quality; write: procurement
GET / POST	/api/shipments, /api/shipments/:id/events	Lists shipments and adds event updates.	PostgreSQL shipments and events	Read: logistics roles; write: procurement/manager
GET / POST	/api/qc-reports	Lists reports and creates a new versioned QC report.	MongoDB qc_reports	Read: permitted roles; write: quality

POST	/api/qc-reports/:id/approve	Approves a QC report.	MongoDB qc_reports	Quality inspector
GET / POST / PATCH	/api/certifications	Lists, creates, or updates pending certifications.	MongoDB certifications	Read: permitted roles; write: quality
POST	/api/certifications/:id/approve	Locks a certification as immutable.	MongoDB certifications	Quality inspector
GET / POST	/api/iot	Lists IoT sensor readings or submits telemetry.	MongoDB iot_sensor_logs	Read/write: equipment engineer
GET / POST	/api/compliance-flags	Lists or raises compliance flags.	MongoDB compliance_flags	Auditor can create flags
GET	/api/audit-logs	Lists recent audit trail events.	PostgreSQL audit_logs	Manager and auditor
GET	/api/export/:collection	Exports permitted datasets as CSV.	PostgreSQL or MongoDB	Manager and auditor

5. Dashboard Evidence

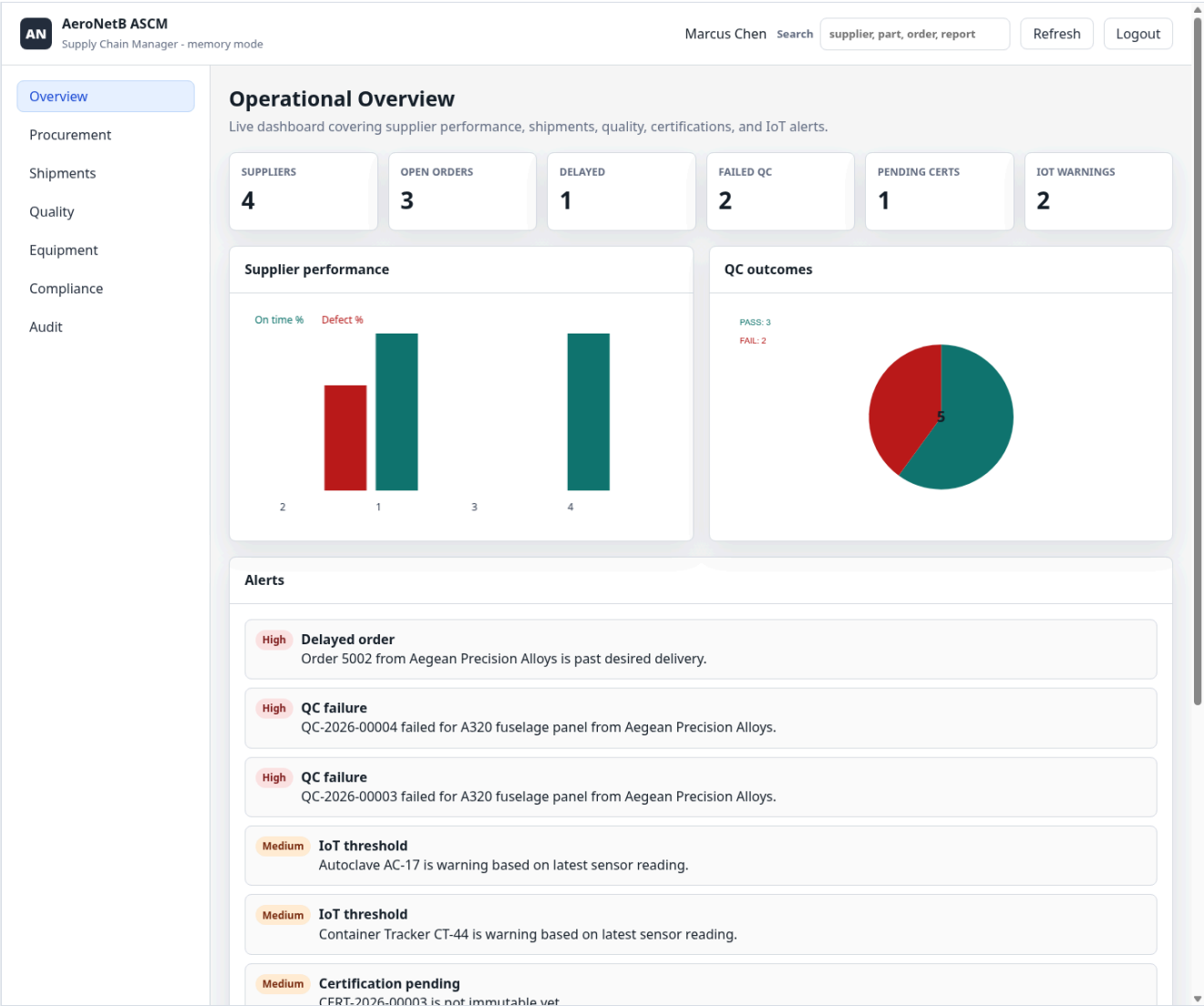


Figure 3 - Supply Chain Manager overview dashboard with KPIs and charts

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Procurement Officer - memory mode

Priya Raman

Search

supplier, part, order, report

Refresh

Logout

Overview

Procurement

Shipments

Quality

Compliance

Procurement

Create suppliers and orders, then track the order lifecycle.

Add supplier

Business name

Accreditation

Contact email

Contact name

Phone

Address

Add supplier

Create purchase order

Supplier

Part

Quantity

Desired delivery

Status

Create order

Suppliers

ID	BUSINESS	ACCREDITATION	CONTACT
2	Aegean Precision Alloys	ISO 9001	Nikos Markou orders@aegean-alloys.example
1	Helios Aero Composites	AS9100, ISO 9001	Klara Weber supply@helios-aero.example
3	NorthStar Avionics Materials	AS9100	Maya Deschamps quality@northstar-materials.example
4	Pacific AeroForge	Pending renewal	Lee Wei Chen sales@pacific-aeroforge.example

Orders

Figure 4 - Procurement view for supplier and order workflow

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Quality Inspector - memory mode

Lena OrtizSearchsupplier, part, order, reportRefreshLogout

Overview

Procurement

Shipments

Quality

Compliance

Quality Control

Create versioned QC reports and lock certifications after approval.

Create QC report

Order

Order 5005

Supplier

Aegean Precision Alloys

Part

A320 fuselage panel

Report type

visual

Outcome

PASS

Notes

inspection result details

Create QC report

Create certification

Supplier

Aegean Precision Alloys

Part

A320 fuselage panel

Batch origin

Test summary

PENDING

Digital signature

SHA256:pending

Create certification

QC reports

REPORT	SUPPLIER	PART	TYPE	OUTCOME	APPROVED	ACTION
QC-2026-00004 v2	Aegean Precision Alloys	A320 fuselage panel	ndt	FAIL	pending	Approve
QC-2026-00005 v1	Pacific AeroForge	Composite landing gear door	environmental	PASS	pending	Approve
QC-2026-00003 v1	Aegean Precision Alloys	A320 fuselage panel	ndt	FAIL	pending	Approve
QC-2026-00002 v1	Aegean Precision Alloys	Titanium fastener kit	visual	PASS	PASS	-
QC-2026-00001	Helios Aero Composites	A320 fuselage panel	dimensional	PASS	PASS	-

Figure 5 - Quality view for QC report and certification workflow

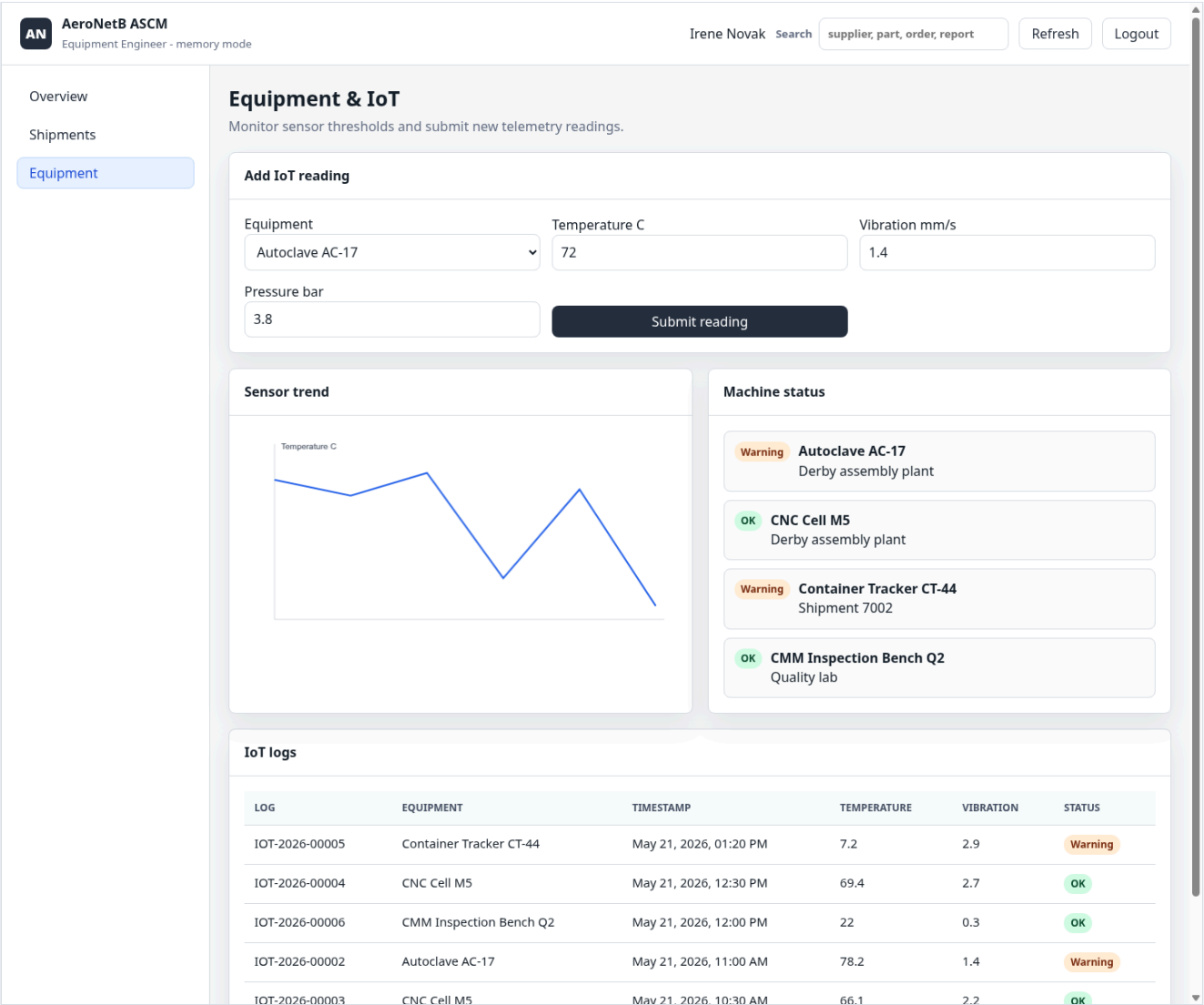


Figure 6 - Equipment engineer IoT monitoring view

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Auditor / Regulator - memory mode

Omar Haddad
Search
supplier, part, order, report
Refresh
Logout

Overview
Procurement
Shipments
Quality
Equipment
Compliance
Audit

Compliance

Export certs

Read-only certification review with auditor flagging for non-compliance cases.

Raise compliance flag

Entity type

Certification

Entity ID

CERT-2026-00003

Severity

Medium

Note

Raise flag

Certification records

CERTIFICATE	SUPPLIER	PART	BATCH	APPROVED	APPROVAL DATE	ACTION
CERT-2026-00003	Pacific AeroForge	Composite landing gear door	Batch-PAF-2026-21	pending	-	-
CERT-2026-00001	Helios Aero Composites	A320 fuselage panel	Batch-HAC-2026-19	completed	May 14, 2026, 01:00 PM	-
CERT-2026-00002	Aegean Precision Alloys	Titanium fastener kit	Batch-APA-2026-44	completed	May 14, 2026, 04:30 PM	-

Compliance flags

FLAG	ENTITY	SEVERITY	NOTE	STATUS
1	Certification CERT-2026-00003	Medium	Seal integrity is still pending before final approval.	open

Figure 7 - Auditor compliance and certification review view

6. Security and Compliance

Control	Implementation
Authentication	Users log in with auth ID and password. Stored passwords are PBKDF2 hashes with salts.
Session protection	The API issues signed HMAC session tokens with expiry.
RBAC	Each endpoint checks permissions in the backend before reading or mutating data.
Auditor restriction	The auditor can read compliance data and raise flags but cannot mutate suppliers, orders, QC reports, or certifications.
Certification immutability	Approved certifications have is_immutable=true. Later update attempts return HTTP 409.
Audit trail	Important actions write a record to audit_logs with emp_id, action type, entity, timestamp, outcome, and details.

7. Testing and Validation

Test	Expected Result	Status
npm run check	JavaScript syntax checks pass for backend and tests.	Passed
npm test	Integration tests pass against a local seeded service.	Passed
Auditor creates supplier	Request is rejected with HTTP 403.	Passed
Procurement creates supplier	Supplier is created and returned with HTTP 201.	Passed
QC report retest	New report receives the next version number.	Passed
Approved certification update	Update is rejected with HTTP 409.	Passed
IoT reading submission	Equipment engineer can submit telemetry.	Passed
CSV export	Manager can export audit logs as CSV.	Passed

8. Deployment Explanation

Render was chosen because it can deploy the Node.js service directly from GitHub and can create a PostgreSQL database from the same Blueprint file. MongoDB Atlas was chosen for document storage because the scenario includes variable QC payloads, certification evidence, and IoT readings. The Render service uses environment variables for all private connection strings. The same codebase runs locally in memory mode when database credentials are not present, which supports offline demonstration and automated testing.

9. Demo Flow

1. Open the Render URL and show `/api/health`.
2. Log in as Supply Chain Manager and show KPIs, alerts, supplier chart, QC chart, search, and CSV export.
3. Log in as Procurement Officer and create a supplier or order.
4. Log in as Quality Inspector and create/approve a QC report, then create and lock a certification.
5. Try to edit the locked certification and show the immutability error.
6. Log in as Equipment Engineer and add an IoT reading.
7. Log in as Auditor and show read-only compliance access plus compliance flag creation.
8. Show the audit log proving actions are attributed to `emp_id`.